

① Find $\frac{df}{dt}$ for the following functions $\leftarrow$ directly chain rule

a) $f(x, y) = \ln(x^2 + y^2)$

$x(t) = t$

$y(t) = t^2$

$t \mapsto (x(t), y(t)) \xrightarrow{f} f(x, y)$

Directly: $f(t) = \ln(t^2 + (t^2)^2)$
 (High School) $\boxed{f'(t) = \frac{df}{dt} = \frac{(t^2 + t^4)'}{t^2 + t^4} = \frac{2t + 4t^3}{t^2 + t^4}}$

Chain rule: 1d: $f(u(t))$

$\boxed{\frac{df}{dt} = \frac{df}{du} \cdot \frac{du}{dt}} = \dots$

$\left(\begin{array}{l} t \rightarrow u \rightarrow f(u) \\ t \rightarrow u(t) \rightarrow f(u(t)) \end{array} \right)$

$(f(u(t)))' = f'(u) \cdot u'$

Chain rule: 2d: $\boxed{\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}}$

$\frac{\partial f}{\partial x} \xrightarrow{y=ct} \frac{(x^2 + y^2)'}{x^2 + y^2} = \frac{2x}{x^2 + y^2}$

$\frac{\partial f}{\partial y} \xrightarrow{x=ct} \frac{(x^2 + y^2)'}{x^2 + y^2} = \frac{2y}{x^2 + y^2}$

partial derivatives
(depend on $>$ variables)

$\frac{dx}{dt} = (x(t))' = t' = 1$

$\frac{dy}{dt} = (y(t))' = (t^2)' = 2t$

$\boxed{\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}} \rightarrow \text{I'm looking at } f \text{ as a function of } t. \downarrow (f(t))'$

$= \frac{2x}{x^2 + y^2} \cdot 1 + \frac{2y}{x^2 + y^2} \cdot 2t \quad \begin{array}{l} x(t) = t \\ y(t) = t^2 \end{array}$

$= \frac{2t}{t^2 + t^4} + \frac{2t^2 \cdot 2t}{t^2 + t^4}$

$$= \frac{4t^3 + 2t}{t^2 + t^4}$$

Chain rule: 3d: $\boxed{\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial f}{\partial z} \cdot \frac{dz}{dt}}$

b) $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$

$x(t) = \cos t$

$y(t) = \sin t$

$z(t) = t > 0$

$t \mapsto (x, y, z) \mapsto f(x, y, z)$

Directly: remove $x, y, z \rightarrow$ rewrite everything only using t

$f(t) = \sqrt{\cos^2 t + \sin^2 t + t^2} = \sqrt{t^2 + 1}$

$\frac{df}{dt} = (\sqrt{t^2 + 1})' = \frac{(1 + t^2)'}{2\sqrt{1 + t^2}} = \frac{2t}{2\sqrt{1 + t^2}} = \frac{t}{\sqrt{1 + t^2}} = f'(t)$

Chain rule: $\boxed{\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial f}{\partial z} \cdot \frac{dz}{dt}}$

$\frac{\partial f}{\partial x} \stackrel{y, z \text{ d.}}{=} \frac{(x^2 + y^2 + z^2)'}{2 \cdot \sqrt{x^2 + y^2 + z^2}} = \frac{2x}{2\sqrt{x^2 + y^2 + z^2}}$

$\frac{\partial f}{\partial y} \stackrel{x, z \text{ d.}}{=} \frac{(x^2 + y^2 + z^2)'}{2 \cdot \sqrt{x^2 + y^2 + z^2}} = \frac{2y}{2\sqrt{x^2 + y^2 + z^2}}$

$\frac{\partial f}{\partial z} \stackrel{x, y \text{ d.}}{=} \frac{2z}{2 \cdot \sqrt{x^2 + y^2 + z^2}}$

$\frac{dx}{dt} = (x(t))' = (\cos t)' = -\sin t$

$\frac{dy}{dt} = (y(t))' = (\sin t)' = \cos t$

$\frac{dz}{dt} = (z(t))' = t' = 1$

$$\boxed{\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial f}{\partial z} \cdot \frac{dz}{dt}}$$

$$= \frac{x}{\sqrt{x^2+y^2+z^2}} \cdot (-\sin t) + \frac{y}{\sqrt{x^2+y^2+z^2}} \cdot \cos t + \frac{z}{\sqrt{x^2+y^2+z^2}} \cdot 1$$

$$\begin{array}{l} x(t) = \cos t \\ y(t) = \sin t \\ z(t) = 1 \end{array} \quad \begin{array}{l} -\cos t \cdot \sin t \\ \cancel{\sqrt{\cos^2 t + \sin^2 t + 1}} \\ \cancel{\sin t \cdot \cos t} \\ \cancel{\sqrt{\sin^2 t + \cos^2 t + 1}} \end{array} + \frac{t}{\sqrt{\sin^2 t + \cos^2 t + 1}}$$

$$= \frac{t}{\sqrt{t^2 + 1}}$$

② $f(u, v, w) = u^2 + v^2 - w$

$$u(x, y, z) = x^2 \cdot y$$

$$v(x, y, z) = y^2$$

$$w(x, y, z) = e^{-xz}$$

Chain Rule

$$(x, y, z) \xrightarrow{\quad} (u, v, w) \xrightarrow{f} f(u, v, w)$$

$$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \text{ in 2 ways.}$$

Directly: $f(x, y, z) = (x^2 y)^2 + (y^2)^2 - e^{-xz}$
 $= \frac{x^4 y^2 + y^4 - e^{-xz}}{1}$

$$\frac{\partial f}{\partial x} \xrightarrow{y, z \text{ ct.}} 4x^3 \cdot y^2 + z \cdot e^{-xz}$$

$$\frac{\partial f}{\partial y} \xrightarrow{x, z \text{ ct.}} 2y \cdot x^4 + 4y^3$$

$$\frac{\partial f}{\partial z} \xrightarrow{x, y \text{ ct.}} x \cdot e^{-xz}$$

Chain rule:

$$\boxed{\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial x}}$$

$$f(u, v, w) = u^2 + v^2 - w \quad | \quad u(x, y, z) = x^2 \cdot y$$

$$\frac{\partial f}{\partial u} \underline{\underline{v, w \text{ ct.}}} \quad 2u.$$

$$\frac{\partial f}{\partial v} \underline{\underline{u, w \text{ ct.}}} \quad 2v.$$

$$\frac{\partial f}{\partial w} \underline{\underline{u, v \text{ ct.}}} \quad -1.$$

$$\frac{\partial u}{\partial x} \underline{\underline{y, z \text{ ct.}}} \quad 2xy$$

$$v(x, y, z) = y^2$$

$$\frac{\partial v}{\partial x} \underline{\underline{y, z \text{ ct.}}} \quad 0$$

$$w(x, y, z) = e^{-xz}$$

$$\frac{\partial w}{\partial x} \underline{\underline{y, z \text{ ct.}}} \quad -z \cdot e^{-xz}$$

$$\begin{aligned} \frac{\partial f}{\partial x} &= 2u \cdot 2xy + 2v \cdot 0 + (-1) \cdot (-z \cdot e^{-xz}) \quad \underline{\underline{u(x, y, z) = x^2 \cdot y}} \\ &= 2 \cdot x^2 y \cdot 2xy + 0 + z \cdot e^{-xz} \\ &= 4x^3 y^2 + z \cdot e^{-xz} \quad (y, z \text{ ct.}) \end{aligned}$$

$$\boxed{\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial y}}$$

$$f(u, v, w) = u^2 + v^2 - w$$

$$\frac{\partial f}{\partial u} \underline{\underline{v, w \text{ ct.}}} \quad 2u.$$

$$\frac{\partial f}{\partial v} \underline{\underline{u, w \text{ ct.}}} \quad 2v.$$

$$\frac{\partial f}{\partial w} \underline{\underline{u, v \text{ ct.}}} \quad -1.$$

$$u(x, y, z) = x^2 \cdot y$$

$$\frac{\partial u}{\partial y} \underline{\underline{x, z \text{ ct.}}} \quad x^2$$

$$v(x, y, z) = y^2$$

$$\frac{\partial v}{\partial y} \underline{\underline{x, z \text{ ct.}}} \quad 2y$$

$$w(x, y, z) = e^{-xz}$$

$$\frac{\partial w}{\partial y} \underline{\underline{x, z \text{ ct.}}} \quad 0.$$

$$\begin{aligned} \frac{\partial f}{\partial y} &= 2u \cdot x^2 + 2v \cdot 2y - 1 \cdot 0 \\ &= 2 \cdot x^2 y \cdot x^2 + 2 \cdot y^2 \cdot 2y \\ &= 2x^4 y + 4y^3 \end{aligned}$$

$$\boxed{\frac{\partial f}{\partial z} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial z} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial z} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial z}}$$

$$f(u, v, w) = u^2 + v^2 - w$$

$$\frac{\partial f}{\partial u} \stackrel{u, w \text{ ct.}}{=} 2u.$$

$$\frac{\partial f}{\partial v} \stackrel{u, w \text{ ct.}}{=} 2v.$$

$$\frac{\partial f}{\partial w} \stackrel{u, v \text{ ct.}}{=} -1.$$

$$u(x, y, z) = x^2 \cdot y$$

$$\frac{\partial u}{\partial z} \stackrel{x, y \text{ ct.}}{=} 0.$$

$$v(x, y, z) = y^2$$

$$\frac{\partial v}{\partial z} \stackrel{x, y \text{ ct.}}{=} 0$$

$$w(x, y, z) = e^{-xz}$$

$$\frac{\partial w}{\partial z} \stackrel{x, y \text{ ct.}}{=} -x \cdot e^{-xz}$$

$$\frac{\partial f}{\partial z} = 2u \cdot 0 + 2v \cdot 0 - 1 \cdot (-z \cdot e^{-xz}) = x \cdot e^{-xz}.$$

Def: **DIRECTIONAL DERIVATIVES.** (derivatives in a certain direction)

$$D_{f_v}(x) = \lim_{h \rightarrow 0} \frac{f(x + h \cdot v) - f(x)}{h} = \underbrace{\nabla f(x) \cdot v}_{\text{scalar prod.}}$$

③. $f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = x^2 + xy$

a) the gradient (∇) and the direction of the steepest descent at $(1, 0)$.

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (2x + y, x)$$

$$\nabla f(1, 0) = (2 \cdot 1 + 0, 1) = (2, 1).$$

Direction of the steepest descent: $-\nabla f(x) = -\nabla f(1, 0) = -(2, 1) = (-2, -1).$

Direction of the steepest ascent: $\nabla f(x) = \nabla f(1, 0) = (2, 1).$

b) The derivative at $(1, 0)$ in the direction of $(1, 1) = e_1 + e_2$.

$$\text{Directional derivative: } \nabla f(1, 0) \cdot (1, 1) = (2, 1) \cdot (1, 1) = 2 \cdot 1 + 1 \cdot 1 = 3.$$

Dot Product: $(a, b) \cdot (c, d) = ac + bd.$

④. $f: \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = \frac{1}{2} \|x\|^2$

a) $\nabla f(x) \rightarrow$ gradient of f at a point x

b) $D_{f_v}(x) \rightarrow$ directional derivative $\begin{matrix} \rightarrow \text{def (lim)} \\ \rightarrow \text{gradient.} \end{matrix}$

$$a) \|x\|^2 = \langle x, x \rangle = x_1^2 + x_2^2 + \dots + x_m^2$$

$$f(x) = \frac{1}{2} (x_1^2 + \dots + x_m^2)$$

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_m} \right) = (x_1, x_2, \dots, x_m)$$

$$\frac{\partial f}{\partial x_1} = \frac{1}{2} \cdot x_1^2 + \frac{1}{2} \cdot x_2^2 + \dots = \frac{1}{2} \cdot 2x_1 = x_1.$$

at. $\rightarrow 0$

...

$$\frac{\partial f}{\partial x_m} = x_m$$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \frac{\partial f}{\partial x_2}(x), \dots, \frac{\partial f}{\partial x_m}(x) \right) = (x_1, \dots, x_m) = x.$$

$$b) D_{f_r}(x) = \nabla f(x) \cdot r = x \cdot r = \langle x, r \rangle$$

scalar prod.
of 2 vectors

$\begin{matrix} > 0 \Rightarrow \nearrow \\ = 0 \Rightarrow \text{neither } \nearrow, \text{ nor } \searrow \\ < 0 \Rightarrow \searrow \end{matrix}$

"Derivative of f in the direction of r "

$$\begin{aligned} D_{f_r}(x) &= \lim_{h \rightarrow 0} \frac{f(x+hr) - f(x)}{h} = \\ &= \lim_{h \rightarrow 0} \frac{\frac{1}{2} \|x+hr\|^2 - \frac{1}{2} \|x\|^2}{h} \\ &= \frac{1}{2} \cdot \lim_{h \rightarrow 0} \frac{\|x+hr\|^2 - \|x\|^2}{h} \\ &= \frac{1}{2} \cdot \lim_{h \rightarrow 0} \frac{\langle x+hr, x+hr \rangle - \langle x, x \rangle}{h} \\ &= \frac{1}{2} \cdot \lim_{h \rightarrow 0} \frac{\cancel{\langle x, x \rangle} + \langle x, hr \rangle + \langle hr, x \rangle + \langle hr, hr \rangle - \cancel{\langle x, x \rangle}}{h} \\ &= \frac{1}{2} \cdot \lim_{h \rightarrow 0} \frac{2\langle x, hr \rangle + \langle hr, hr \rangle}{h} \\ &= \frac{1}{2} \cdot \lim_{h \rightarrow 0} \frac{2h\langle x, r \rangle + h \cdot h\langle r, r \rangle}{h} \\ &= \frac{1}{2} \cdot \lim_{h \rightarrow 0} \frac{h(2\langle x, r \rangle + h\langle r, r \rangle)}{h} \\ &= \frac{1}{2} \cdot 2\langle x, r \rangle + \frac{1}{2} \cdot \lim_{h \rightarrow 0} h\langle r, r \rangle \\ &= \langle x, r \rangle + 0 \end{aligned}$$

$$= \langle x, r \rangle$$

2nd order derivatives:

the variable which is closer to the function f will be evaluated first.

because:

$$\frac{\partial^2 f}{\partial x_j \partial x_i} \neq \frac{\partial^2 f}{\partial x_i \partial x_j} \quad (\text{in general})$$

if $= \Rightarrow$ symmetric

HESSIAN MATRIX: (SYMMETRIC)

$$H(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_m} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_m \partial x_1} & \frac{\partial^2 f}{\partial x_m \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_m \partial x_m} \end{bmatrix}$$

$$\textcircled{5} D = \text{diag}(d_1, \dots, d_m) = \begin{bmatrix} d_1 & \dots & 0 \\ 0 & d_2 & \vdots \\ \vdots & \vdots & \ddots \\ 0 & \dots & d_m \end{bmatrix} \quad \begin{aligned} \nabla f(x) &\stackrel{?}{=} D x \\ H(x) &\stackrel{?}{=} D \end{aligned}$$

$$f: \mathbb{R}^m \rightarrow \mathbb{R}, f(x) = \frac{1}{2} x^T D x$$

transposed vector (column $\rightarrow$ row
row $\rightarrow$ column)

$$(1 \ 2 \ 3 \ 4) \rightarrow \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}_{m \times 1}$$

$$x^T = [x_1 \dots x_m]_{1 \times m}$$

$$f(x) = \frac{1}{2} \cdot [x_1 \dots x_m] \cdot \begin{bmatrix} d_1 & \dots & 0 \\ \vdots & d_2 & \vdots \\ 0 & \dots & d_m \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$$

$$= \frac{1}{2} \cdot [x_1 \dots x_m] \cdot \begin{bmatrix} d_1 x_1 \\ \vdots \\ d_m x_m \end{bmatrix}$$

$$= \frac{1}{2} (x_1^2 \cdot d_1 + x_2^2 \cdot d_2 + \dots + x_m^2 \cdot d_m)$$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \frac{\partial f}{\partial x_2}(x), \dots, \frac{\partial f}{\partial x_m}(x) \right)$$

$$\frac{\partial f}{\partial x_1} = \frac{1}{2} \cdot x_1^2 \cdot d_1$$

$$= (d_1 x_1, d_2 x_2, \dots, d_m x_m)$$

$$= D \cdot x \text{ linear function}$$

$$= \frac{1}{x} \cdot d x_1 \cdot d_1$$

$$= x_1 \cdot d_1$$

$$H(x) = ?$$

! The derivative is a vector.

$$\frac{\partial^2 f}{\partial x_1 \partial x_1} = \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) \stackrel{d_1 d_1}{=} \frac{\partial}{\partial x_1} (x_1 d_1) = d_1 \checkmark$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_2} \right) = \frac{\partial}{\partial x_1} (x_2 d_2) \stackrel{\text{no } x_1}{=} 0$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_j} = \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_j} \right) = \frac{\partial}{\partial x_1} (d_j x_j) = 0 \quad j \neq 1$$

$$\frac{\partial^2 f}{\partial x_2 \partial x_2} = \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_2} \right) = \frac{\partial}{\partial x_2} (d_2 x_2) = d_2$$

$$\text{So: } \frac{\partial^2 f}{\partial x_i \partial x_j} = 0, j \neq i$$

$$H(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_m} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_m \partial x_1} & \frac{\partial^2 f}{\partial x_m \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_m \partial x_m} \end{bmatrix} = \begin{bmatrix} d_1 & \dots & 0 \\ \vdots & d_2 & \ddots \\ \vdots & \vdots & \ddots & d_m \end{bmatrix} = 0 \text{ constant matrix.}$$

